

$$\text{rank}(A) = r$$

$$P^{-1}AP = D$$

$$A\mathbf{x} = \mathbf{b}$$

$$A\mathbf{v} = \lambda\mathbf{v}$$

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$$\text{tr}(A) = \sum \lambda_i$$

$$\delta_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \dots$$

$$\det(A - \lambda I) = 0$$

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$$\text{tr}(A) = \sum \lambda_i$$

$$A^{-1} = \frac{1}{\det A} \text{adj}(A)$$

$$\left(\begin{array}{ccc|ccc} 1 & & & & & \\ 2 & 0 & & 1 & & \\ 3 & 1 & 1 & & & \\ 2 & & 1 & & & \\ & 2 & & & & \end{array} \right) \begin{array}{l} \\ \\ \\ \\ 9 \end{array}$$

$$A^{-1} = \frac{1}{\det A} \text{adj}(A)$$

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